Setting up SFML in Visual Studio 2022

This document describes how to get set up with SFML (Simple Fast Multimedia Library) in Visual Studio 2022. The result should be essentially "Hello World" for the SFML library.

This lab was written for SFML 2.5.1. SFML 3.0 was released in the last year (a major version upgrade)– with some changes to events and object creation that require re-writing the documentation. So, for this iteration, we will use the older version: 2.5.1.

Follow the instructions below ***carefully*** and you should only need to do this once.

Steps:

Go to http://sfml-dev.org

Click the Download tab, and click "Older Versions", and then 2.5.1



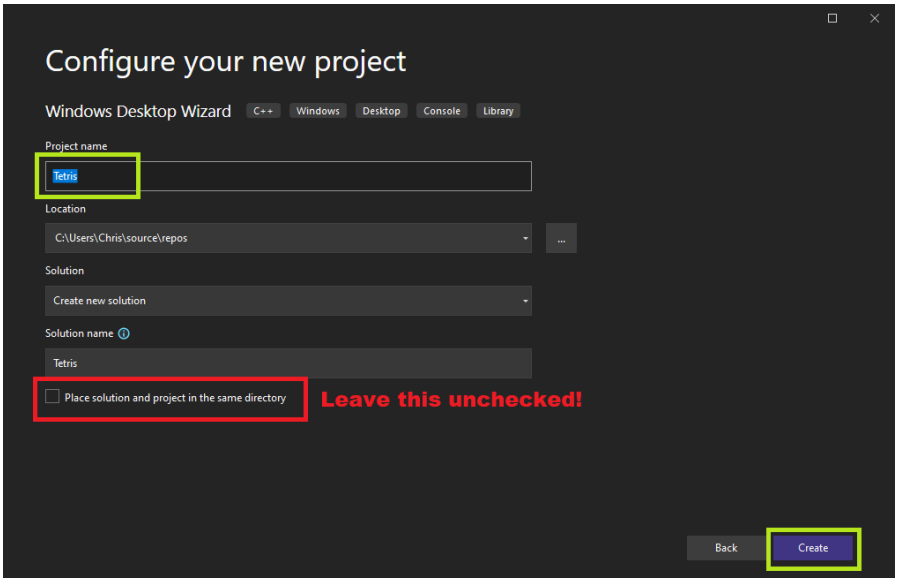
Select the download button for Visual C++ 15 (2017) - 32-bit (32 bit is for the platform you want to build for, not just the computer you are on).

Unzip the file to a folder on your desktop (probably called SFML-2.5.1)

Create an empty Tetris project (targeting a 32 bit OS):

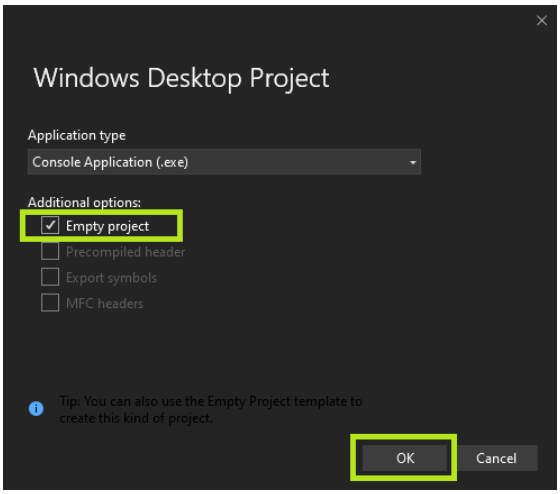
In Visual Studio 2022: **File | New | Project**

Select **Windows Desktop Wizard** and click **Next**.

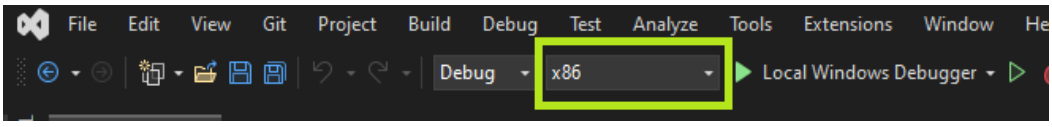


Enter a Name for the project: "Tetris". Ensure "**Place solution and project in the same directory**" is **NOT** checked. Click **Create**.

Check the box for **Empty Project** (below), and click **OK**.



In the toolbar, set the target platform to **x86** (a 32 bit OS) using the dropdown shown below. This will allow our executable to run on both 32 and 64 bit systems (instead of limiting it to just 64 bit systems).



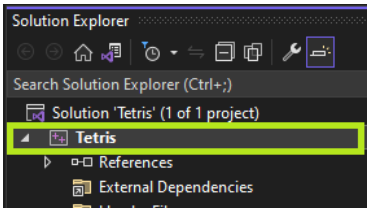
Copy SFML files into the project:

In the Solution Explorer, highlight the name of the project ("Tetris", just below the Solution), right click, and then left click on "**Open Folder in File Explorer**".

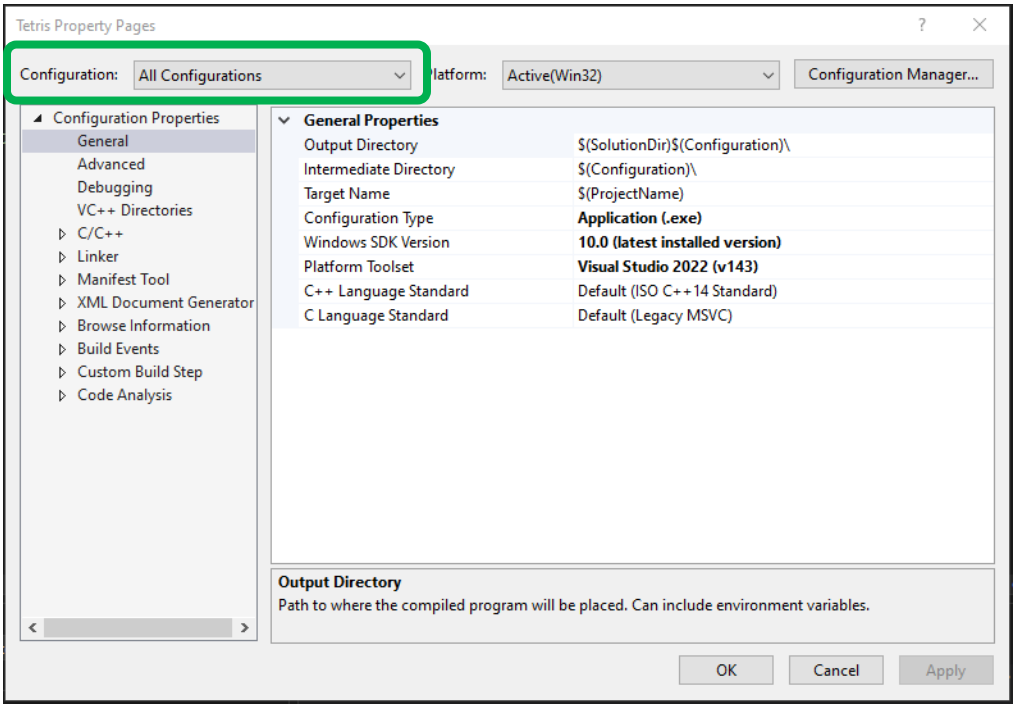
Navigate one folder back to the folder that the Tetris.sln file is stored in. (it is also called Tetris). Create a new folder here called "**SFML**"

Go to the folder on your desktop that the SFML files are in. Copy the bin, include, and lib directories and paste them into the SFML folder you created above.

Configure the project (All Configurations):



Return to Visual Studio, right click on the project name in the Solution Explorer, and left click on Properties. The following window (below) should come up.



Change the Configuration dropdown box to "**All Configurations**" (both Debug and Release).

Highlight: **Configuration Properties | Linker | General | Additional Library Directories**.

Click the dropdown to allow you to edit the fields, select Edit.

Click the "New Line" icon at the top, click on the field to edit it, and enter **"..\SFML\lib".** Click OK and Apply.

SFML Path Update
Some have reported difficulty with this using this path doesn't work for them. This is likely due to an option in Visual Studio 2022 (when you create a new project) to "Place solution and project in same directory". If this was checked when you created your project, you will likely need to use a slightly

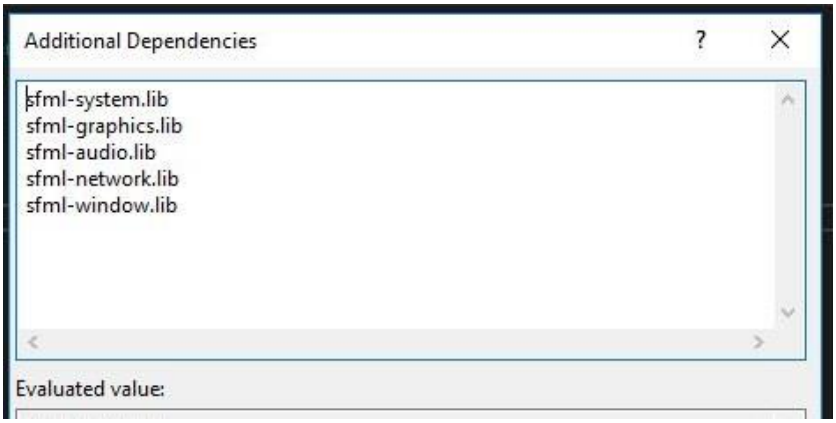
different path: **".\SFML\lib"** (with a single period).

The "..\" prefix means look in the previous folder, whereas ".\" means look in the current folder. If your SFML folder is located in the parent folder relative to your source files, then "**..\SFML\lib**" should work. If it is in the same folder, then use "**.\SFML\lib**". Whether it worked can be determined later on. In a later step, you will be able to tell if the path is correct. VS2022 should trigger an expected error message (see "*Figure 1: Expected Error Message*" shown 2 pages below).

Highlight: **Configuration Properties | Linker | Input | Additional Dependencies**. Click on the dropdown & select edit.

An Additional Dependencies window will pop up. For SFML we need to include 5 different libraries in this particular order: sfml-system.lib, sfml-graphics.lib, sfml-audio.lib, sfml-network.lib, sfml-window.lib

Click **OK** (the Additional Dependencies dialog) and then click **Apply**.



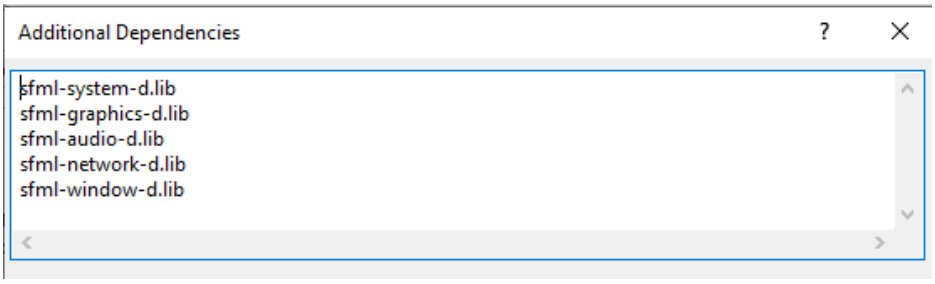
Configure the project (Debug Configuration):

We'll now modify the configuration for the Debug version:

Change the Configuration from "**All Configurations**" to "**Debug**", because the debug version uses different lib files.

Highlight: **Configuration Properties | Linker | Input | Additional Dependencies**. Click on the dropdown & select edit.

Edit the files that are there to add a -d just before the .lib extension.



Click **OK** & **Apply**, and then **OK** to close the Properties dialog.

Add some source code for main():

Left click on Source Files in the Solution Explorer, and select **Add | New Item**.

Select C++ File (.cpp) and click Add (to add Source.cpp).

Cut and paste the following code (below) into the Source.cpp file. (this was copied directly from the sfml tutorial webpage: https://www.sfml-dev.org/tutorials/2.5/start-vc.php)

Note: D2L doesn't allow you to copy properly, so do this after downloading the document from D2L, or better yet, copy directly from the webpage.

```
#include <SFML/Graphics.hpp>

int main()
{
    sf::RenderWindow window(sf::VideoMode(200, 200), "SFML works!");
    sf::CircleShape shape(100.f);
    shape.setFillColor(sf::Color::Green);

    while (window.isOpen())
    {
        sf::Event event;
        while (window.pollEvent(event))
        {
            if (event.type == sf::Event::Closed)
                window.close();
        }

        window.clear();
        window.draw(shape);
        window.display();
    }

    return 0;
}
```

Save the file.

The code is not expected to work quite yet. Keep going!

Continue configuring the project (All Configuration):

Right click again on the project name in the Solution Explorer, and left click on Properties. Note that a new category (C/C++) has appeared under Configuration Properties

Switch the Property Pages Configurations back to "**All Configurations**" so we're not just affecting the Debug or Release version.

Highlight **Configuration Properties | C/C++ | General | Additional Include Directories**. Edit it, add in the sfml\include directory (ie: **..\SFML\include**). Click **OK** and **Apply**. Click **OK** to close this window.

Attempt to build the project. It will fail with the following error (see Figure 1), but it will create a Debug folder in your project folder.

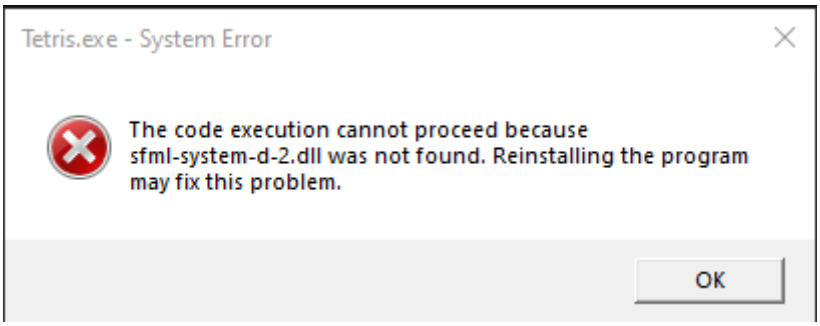


Figure 1: Expected Error Message

If this error message does not happen, please see the SFML Path Update highlighted earlier in this documentation. Please let me know (newsteadc@camosun.bc.ca) if this does not resolve the issue.

In windows explorer, go to the sfml/bin folder. Select all the files that end with a "-d", along with the openal32.dll file. Copy these, and then paste them in your /Tetris/Debug folder. (note: careful not to confuse this with the Debug folder created one folder down in your project folder. The correct folder should also contain an exe file)

Test it out!

Try compiling and running. If everything went well, you should see a square window with a green circle in it appear.

If this happens… rejoice! It means you have set up SFML correctly and and ready to start coding!

If it doesn't (hopefully not the case!), it means there may have been a step missed. Go back and try to figure out if everything was done correctly.

